BARCLAYS HACK-O-HIRE · 2026

Enterprise AI Compliance System · Technical Submission

SAR Narrative Generator

with Intelligent Audit Trail

Evidence-Bounded Generation · Deterministic AML Logic · Regulator-Replayable Auditability

Team	Dominos
Track	SAR Narrative Generator with Audit Trail

1. Abstract

~75%	£3–5M	100%	Zero
Reduction in analyst effort per SAR	**Estimated annual saving (10K SARs)**	**Automated audit trail coverage**	**PII exposure to the LLM — by design**

Financial institutions file hundreds of thousands of Suspicious Activity Reports annually, each demanding five to six hours of manual analyst effort to collect fragmented evidence, interpret transactional patterns, and produce a regulator-compliant narrative. The cumulative cost is measured in millions of pounds. The compliance risk from inconsistent output is material.

The SAR Narrative Generator eliminates this burden through a purpose-built hybrid pipeline: a deterministic AML rule engine makes the compliance decision; a Retrieval-Augmented Generation layer grounds the output in approved regulatory precedents; a constrained LLM converts structured evidence into regulator-aligned prose. Critically, raw transaction data and PII never reach the AI model — this boundary is enforced architecturally, not by policy. Every sentence is traceable to a specific rule trigger. Every case generates a regulator-replayable audit record. Every submission requires mandatory human analyst approval.

The result is a system that drafts compliant SARs in under ninety minutes, produces consistent regulatory-quality narratives across all analysts, and generates the complete evidentiary audit trail that regulators increasingly demand — at enterprise scale, with deployable security controls.

2. Problem & Industry Impact

The UK's National Crime Agency received over 900,000 SAR submissions in 2022–23. Each represents a compliance analyst manually aggregating evidence across four to six siloed systems, interpreting transactional patterns, and producing a structured narrative under time pressure. The operational cost is unsustainable. The quality variance is a regulatory liability.

Industry Pain Point	Regulatory & Business Consequence
5–6 analyst hours per SAR	Investigation backlogs during high-alert periods; analyst fatigue and attrition
Narrative inconsistency	Quality variance by analyst skill; inconsistent SARs invite regulatory scrutiny
Fragmented evidence access	4–6 disconnected systems; incomplete evidence leads to narrative gaps and resubmission cost
No structured audit trail	Cannot demonstrate reasoning provenance; elevated enforcement risk during regulatory examination
Generic AI tools unsuitable	Hallucination risk, PII exposure, no audit trail, no regulatory schema compliance

MARKET GAP

The industry needs a system that automates evidence assembly, applies deterministic compliance logic, generates schema-enforced narratives from verified data only, and produces a complete regulator-replayable audit trail — within a mandatory human governance framework.

3. Solution Overview

The SAR Narrative Generator is an AI-assisted compliance platform that transforms SAR drafting from a five-hour manual process into a supervised thirty-minute workflow. It is built on three non-negotiable design principles:

- Separation of decision logic from narrative generation: compliance decisions are made deterministically; the LLM only composes prose from pre-validated evidence.
- Evidence-bounded generation: the LLM operates within a closed, sanitised information space — it cannot speculate beyond what the rule engine has verified.
- Complete, regulator-replayable auditability: every component action, model configuration, and analyst decision is logged immutably and reproducible on demand.

System Pillar	What It Does	Why It Matters
Deterministic Rule Engine	YAML-defined AML rules evaluate transaction patterns and generate structured evidence blocks	Compliance decisions are fully reproducible, auditable, and jurisdiction-configurable

Evidence Pack Builder	Strips PII and raw data; passes only anonymised evidence summaries to the LLM	Architectural guarantee: zero PII reaches the AI layer — not configurable, not bypassable
RAG Retrieval	Retrieves top-5 semantically similar approved SARs and regulatory guidance from ChromaDB	Grounds LLM output in institutional precedent; eliminates typological drift
LLM Engine (Llama 3.1)	Generates structured JSON narrative across schema-enforced regulatory fields at temp 0.3	Controlled, evidence-constrained generation with post-generation validation gate
Audit Trail	Immutable PostgreSQL record of every rule, evidence block, prompt version, model config, and analyst action	Full replay capability for SR 11-7 / CP 6/22 model governance and regulatory examination

4. Core Innovation

4.1 The Hybrid Architecture — Why Three Layers, Not One

The fundamental innovation is not the use of an LLM — it is the precise, enforced separation of three layers that each address the failure modes of the others:

Layer	Unique Capability	Why It Cannot Work Alone
Deterministic Rule Engine	Makes exact, reproducible compliance decisions from versioned AML logic	Produces structured flags — not human-readable narrative. Cannot synthesise prose.
RAG Retrieval (ChromaDB)	Grounds generation in approved regulatory precedents and typology examples	Cannot evaluate raw transaction data or make case-specific compliance determinations
LLM (Llama 3.1, temp 0.3)	Converts structured evidence into fluent, regulator-aligned prose	Without grounding, hallucinates facts; cannot make compliance determinations

INNOVATION CORE

The rule engine's determinism eliminates the LLM's greatest weakness — factual invention. The RAG layer eliminates its second weakness — typological drift. The LLM eliminates the rule engine's only limitation — inability to produce readable prose. Combined, they achieve what no single technology can deliver alone.

4.2 PII Isolation Architecture

The Evidence Pack Builder enforces a one-way, lossy transformation between the Decision Layer and the Narrative Layer. This is not a filter or a configuration setting — it is an architectural boundary. The LLM never receives account numbers, customer identifiers, raw transaction lists, or any PII. It receives only an anonymised evidence pack: pattern summaries, confidence-scored indicators, and retrieved regulatory excerpts. This boundary is the primary hallucination control mechanism. The LLM cannot invent figures it was not given.

4.3 Sentence-Level Audit Traceability

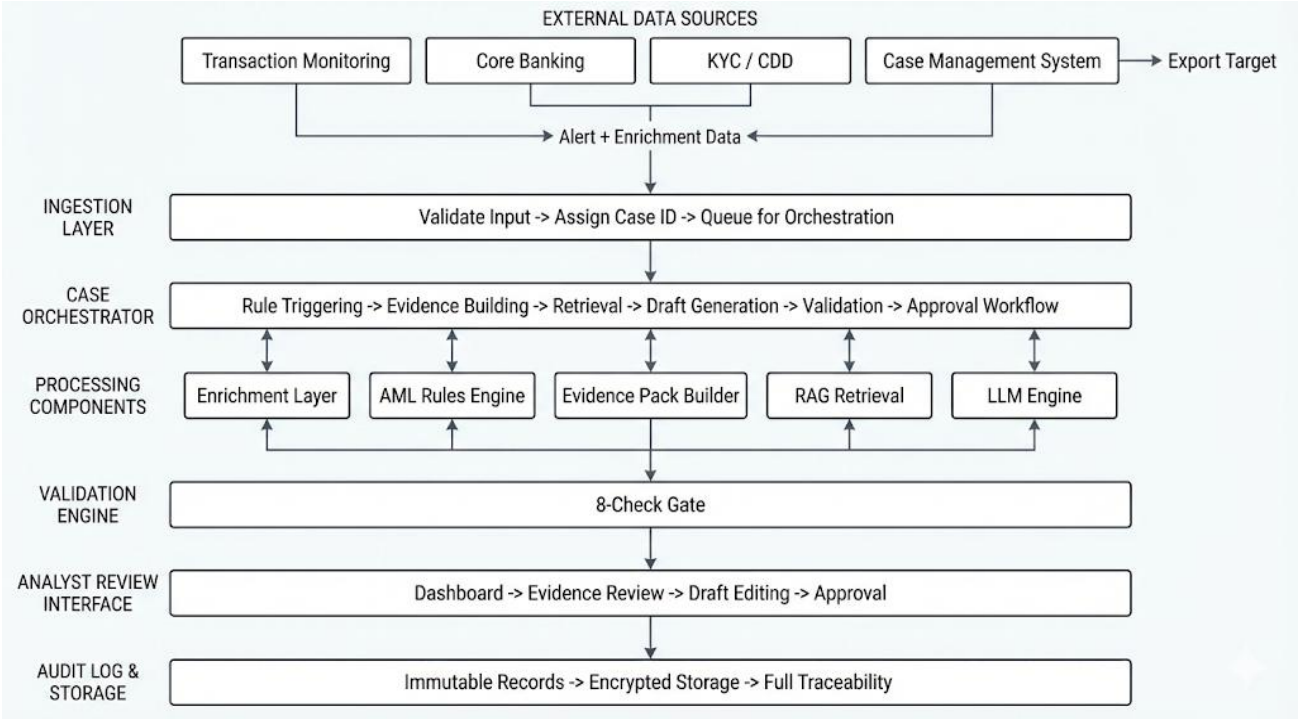
Every sentence in the generated narrative is linked to the specific evidence elements that informed it — rule triggers, transaction indicators, and retrieved precedent excerpts. These links are surfaced in the analyst review interface and stored in the immutable audit log. Each sentence receives an automated explainability score; low-scoring sentences are flagged and cannot be submitted without explicit analyst acknowledgement.

```
SENTENCE-LEVEL TRACEABILITY EXAMPLE
──────────────────────────────────────────────────────────────────────
Sentence: "The subject executed 47 transfers totalling ₹50,00,000 within 7 days."
┌─ Evidence Links ──────────────────────────────────────────────────
│  Rule:      AML-001 [Structuring] — confidence 0.82 — evidence §1-3
│  Txns:      TXN-4582 through TXN-4628 (aggregated in enrichment layer)
│  Typology:  SAR-REF-0447 [Layering — Cross-Border Aggregation]
│  Score:     Explainability 0.94 — PASS
└──────────────────────────────────────────────────────────────────
```

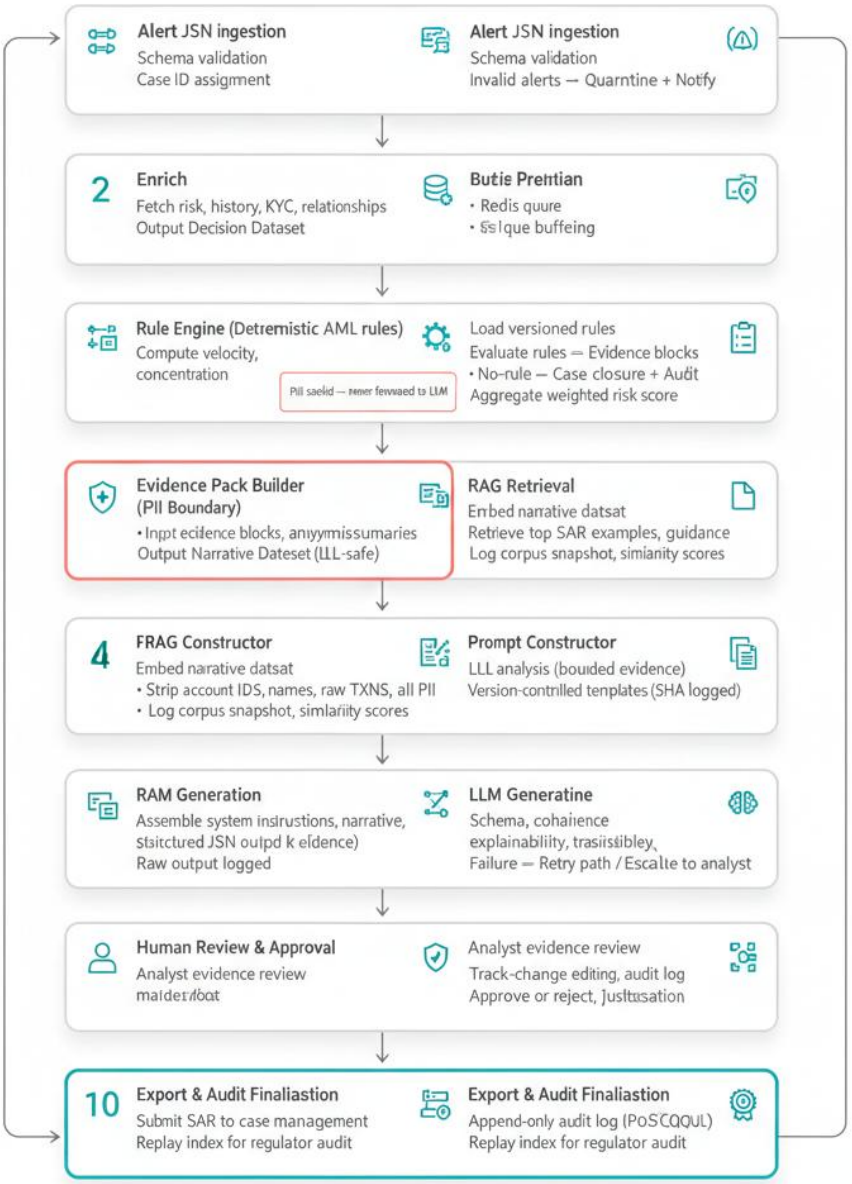
4.4 Regulator-Replayable Audit Trail

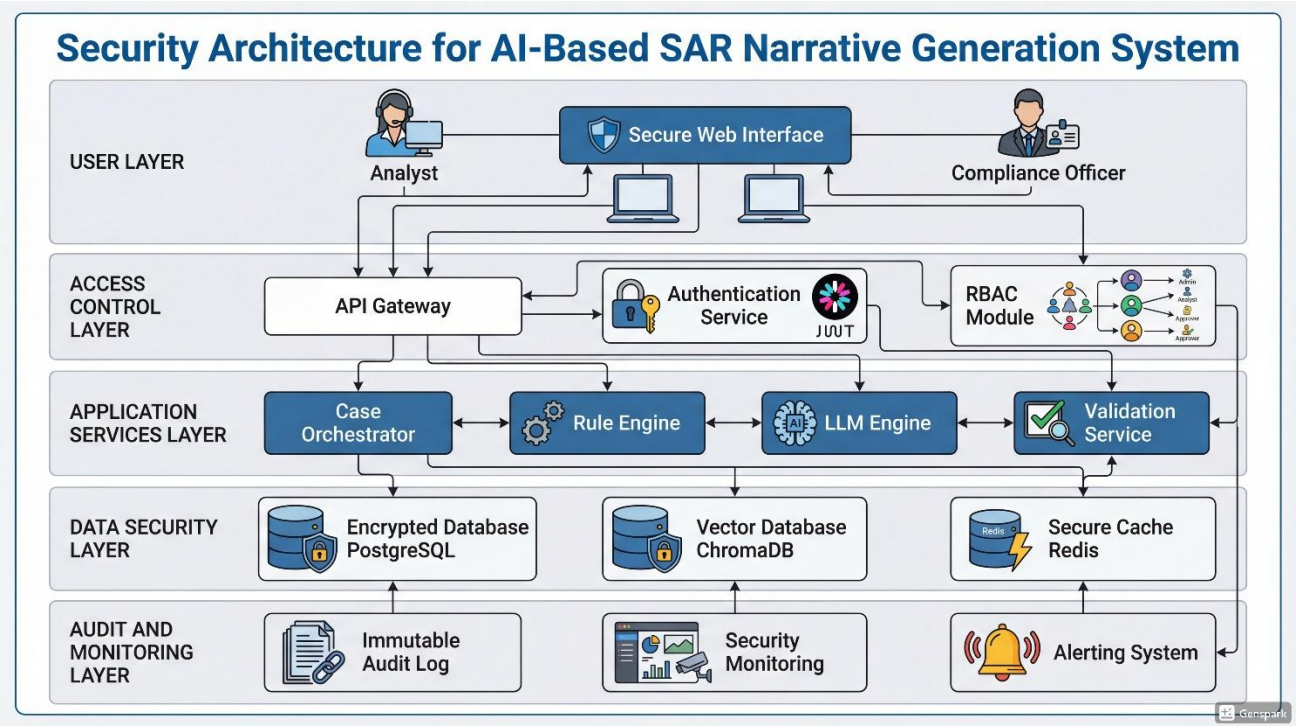
Any filed SAR can be fully replayed on demand. The system stores: the sealed evidence pack, the exact Jinja2 prompt template version (SHA-pinned), the model name and inference parameters, and the ChromaDB corpus snapshot ID. Given a case ID, the orchestrator restores this state exactly and regenerates the draft deterministically. Replay output is automatically compared to the original; any divergence triggers a model risk alert. This satisfies SR 11-7, CP 6/22, and FCA model governance expectations for AI systems in financial services.

5. System Architecture

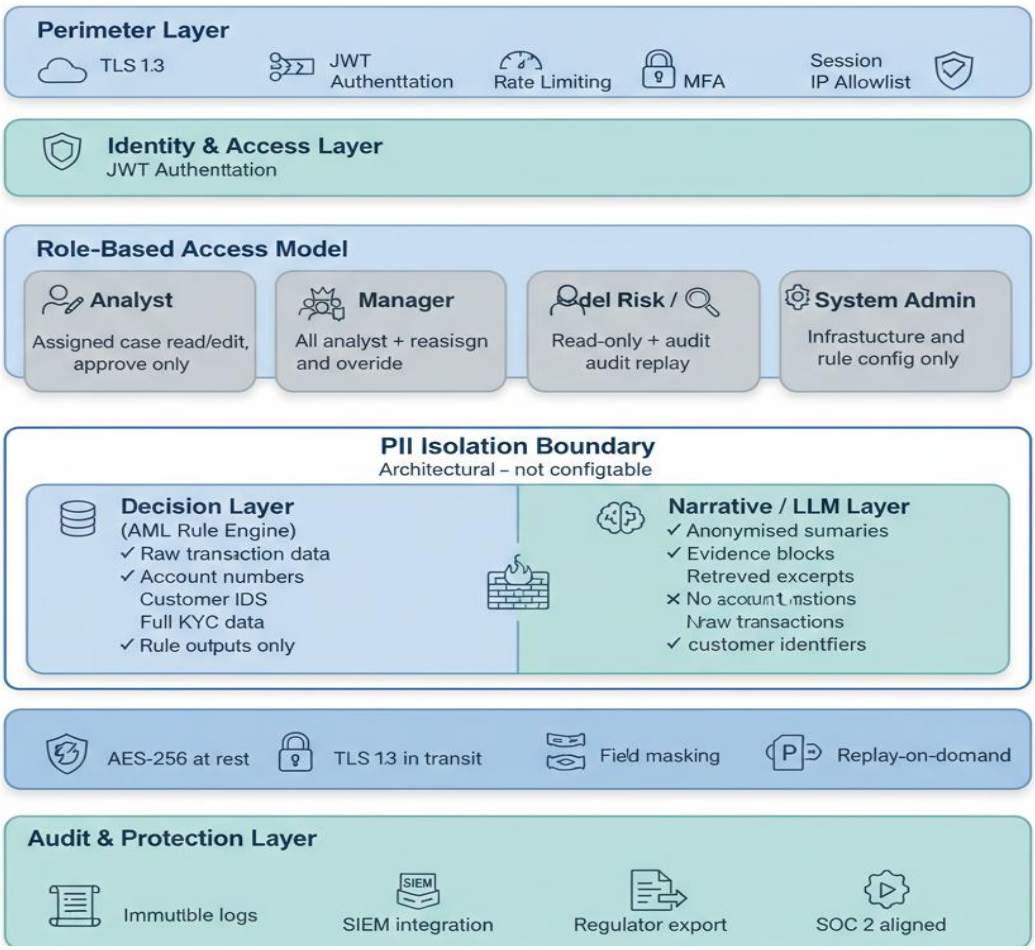

EXTERNAL DATA SOURCES
Transaction Monitoring
Core Banking
KYC / CDD
Case Management System
Export Target
Alert + Enrichment Data
INGESTION LAYER
Validate Input -> Assign Case ID -> Queue for Orchestration
CASE ORCHESTRATOR
Rule Triggering -> Evidence Building -> Retrieval -> Draft Generation -> Validation -> Approval Workflow
PROCESSING COMPONENTS
Enrichment Layer
AML Rules Engine
Evidence Pack Builder
RAG Retrieval
LLM Engine
VALIDATION ENGINE
8-Check Gate
ANALYST REVIEW INTERFACE
Dashboard -> Evidence Review -> Draft Editing -> Approval
AUDIT LOG & STORAGE
Immutable Records -> Encrypted Storage -> Full Traceability

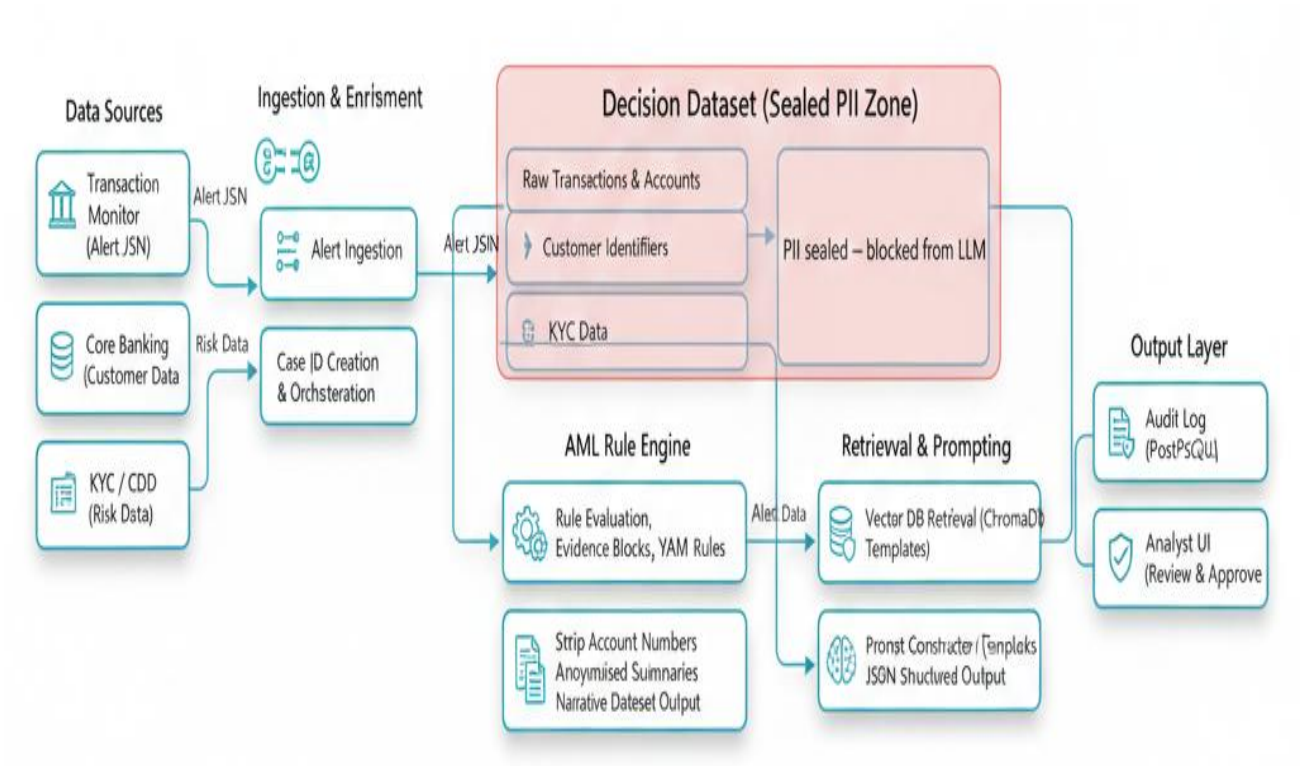
AML SAR Processing Workflow — From Alert Ingestion to Approved Submssion




Security Architecture for AI-Based SAR Narrative Generation System
USER LAYER
Analyst
Secure Web Interface
Compliance Officer
ACCESS CONTROL LAYER
API Gateway
Authentication Service
JWT
RBAC Module
Admin
Analyst
Approver
Approver
APPLICATION SERVICES LAYER
Case Orchestrator
Rule Engine
LLM Engine
Validation Service
DATA SECURITY LAYER
Encrypted Database PostgreSQL
Vector Database ChromaDB
Redis
Secure Cache Redis
AUDIT AND MONITORING LAYER
Immutable Audit Log
Security Monitoring
Alerting System
Genspark

Security & Compliance Boundary — Defence-in-depth Architecture





6. Methodology

6.1 Scalability Design

Every processing component is stateless with respect to case data — all state is persisted in PostgreSQL and Redis. This enables horizontal scaling by adding instances without architectural modification. The Case Orchestrator distributes work through Redis queues, allowing concurrent processing across enrichment, rule engine, and LLM inference workers.

Component	Scaling Mechanism	Enterprise Target
Ingestion (FastAPI)	Kubernetes HPA — horizontal pod autoscaling	Thousands of alerts/minute via Kafka integration
Rule Engine	Stateless, embarrassingly parallel	Sub-second per case on standard compute
LLM Inference	vLLM multi-GPU batch inference, distributed nodes	~3 min per narrative; parallelisable across GPU fleet
Vector Retrieval	ChromaDB distributed with index sharding	Millisecond retrieval at enterprise corpus scale
PostgreSQL (Audit)	Read replicas + partitioned tables by month	10M+ audit records with sustained query performance

6.2 Performance Optimisation

- Risk score computed during rule evaluation to prioritise the analyst queue — highest-risk cases surface first without additional processing.
- ChromaDB HNSW indexing ensures sub-50ms retrieval for the RAG layer regardless of corpus size.
- LLM generation uses structured JSON schema output — reduces token waste and eliminates post-generation parsing overhead.
- Validation engine runs rule checks in parallel using Python multiprocessing — the 8-check gate adds under 2 seconds to total processing time.
- Redis caching for frequently accessed enrichment data (customer risk profiles, static KYC attributes) reduces database round-trips by ~60% at steady state.

6.3 Security Architecture

Defence-in-depth is implemented across five independent layers: network perimeter (TLS 1.3, API gateway, rate limiting), identity (JWT, RBAC, MFA), data (AES-256 at rest, field-level PII masking in audit logs), the architectural PII isolation boundary (Evidence Pack Builder), and the immutable audit compliance layer (append-only PostgreSQL JSONB, SIEM integration). No single layer failure compromises the complete security posture.

6.4 Responsible AI Controls

Control	Mechanism and Justification
Hallucination prevention	LLM receives only pre-validated evidence; temperature 0.3; post-generation fact-consistency validation gate
Bias prevention	Deterministic rule engine (not ML-learned); corpus bias review before inclusion; prohibited keyword scan on all outputs
Prompt injection protection	Jinja2 strict input sanitisation; no user-derived strings interpolated directly; system instructions immutable at runtime
PII leakage prevention	Architectural Decision/Narrative layer separation; Evidence Pack Builder enforces one-way lossy transformation
Model drift detection	5 tracked KPIs: validation failure rate, analyst edit frequency, rejection patterns, explainability score distribution, retrieval relevance
Corpus governance	All RAG corpus documents are approved, anonymised, version-controlled; snapshot IDs logged per retrieval for reproducibility

6.5 Evidence Block Schema

```
{
  "rule_id":          "AML-001",
  "rule_name":        "Structuring — Sub-Threshold Aggregation",
  "confidence_score": 0.82,
  "evidence": [
    "47 transactions in 7-day window  (alert threshold: 20)",
    "Avg amount ₹95,000  (reporting threshold: ₹100,000)",
    "47 unique counterparties — no repeat sources",
    "Consolidated outflow within 4 hrs of final inbound receipt"
```

```
    ],
    "rule_version": "v2.3.1",
    "triggered_at": "2026-03-10T14:32:07Z"
  }
```

7. Audit Trail & Explainability

The audit trail is the system's primary mechanism of regulatory accountability. Every element is stored in PostgreSQL JSONB, encrypted at rest, written as an append-only record. No audit entry can be modified or deleted after creation.

Audit Record Element	Regulatory Purpose
Ingestion timestamp + masked alert data	Establishes event trigger and complete timeline provenance
Triggered rules + confidence scores + evidence blocks	Documents the complete deterministic compliance logic and its outputs
Retrieved doc IDs + similarity scores + corpus snapshot ID	Enables exact retrieval reproduction at any future date for replay
Prompt template version + SHA hash	Guarantees prompt construction is precisely verifiable and reproducible
Model name, version, inference parameters	Satisfies SR 11-7 / CP 6/22 model governance reproducibility requirements
Raw LLM JSON output (pre-processing snapshot)	Distinguishes AI-generated content from post-processing and analyst edits
Analyst edits — section-level diffs with ID and timestamp	Precisely documents all human modifications for accountability and AI/human distinction
Approval decision + mandatory structured comment	Definitive record of human judgment; no approval possible without this entry

REPLAY CAPABILITY

Given any Case ID, the system restores the original evidence pack, loads the SHA-pinned prompt template, applies the stored model configuration, and regenerates the draft. Replay output is automatically compared to original; divergence triggers a model risk alert. Regulators can examine the complete reasoning chain for any filed SAR at any point in the future.

8. Human Analyst Workflow

The analyst is not a rubber stamp. They are the accountable authority over every submission. The system eliminates evidence assembly and first-draft composition, directing analyst cognitive capacity to investigation quality, contextual interpretation, and regulatory judgment. No submission pathway exists without explicit analyst approval.

Step	Analyst Action	System Support
①	**Case assignment**	Risk-scored dashboard surfaces highest-priority cases first; SLA timer active
②	**Evidence review**	Evidence panel displays rule triggers, confidence scores, and source indicators side-by-side with draft
③	**Narrative review and edit**	Track-changes editor; flagged sentences highlighted with explainability score; section-level structure preserved
④	**Source verification**	Masked source data viewer allows evidence verification without exposing PII in the UI layer
⑤	**Approve or reject**	Mandatory structured comment required for both decisions; comment captured immutably in audit log
⑥	**Export trigger**	Orchestrator initiates export to Case Management System; success/failure event logged; audit record sealed

9. Performance Impact

Metric	Baseline (Manual)	With This System	Impact
Analyst effort per SAR	5–6 hours	60–90 minutes	**↓ ~75%**
Automated draft generation time	N/A	< 3 minutes	**New cap.**
Evidence-to-text accuracy	Analyst-variable	~92% validated	**Consistent**
Schema validation pass rate (first pass)	N/A	~95%	**High**
Audit trail completeness	Partial / manual	100% automated	**Complete**
Narrative consistency across analysts	High variance	Schema-enforced	**↑ Significant**
Regulatory query response time	Hours–Days	Minutes (replay)	**Near-instant**
SAR backlog reduction (10K/yr institution)	Baseline	Est. 40–50%	**↓ Significant**

10. Business Value

10.1 Cost Model — Mid-Size UK Institution (10,000 SARs / Year)

Assumptions: blended analyst cost £80,000/yr (£40/hr); average 5.5 hrs/SAR currently; 75% effort reduction achievable with the system.

Cost Component	Current Annual (£)	With System (£ est.)
Analyst drafting (10,000 × 5.5 hrs × £40)	2,200,000	550,000
Regulatory query response + audit assembly	300,000	50,000
SAR rejection and resubmission	150,000	30,000
System infrastructure + maintenance	—	200,000
NET ESTIMATED ANNUAL SAVING	**£2,650,000**	**~£1,820,000**

10.2 Non-Financial Value

- Regulatory relationship quality: consistently structured, evidence-grounded SARs improve the quality of financial intelligence available to law enforcement.
- Analyst retention: eliminating repetitive documentation burden improves job quality in a skills-scarce compliance talent market.
- Enforcement risk reduction: the probability of regulatory action resulting from inadequate SAR documentation is materially reduced.
- AI governance leadership: institutions demonstrating mature, explainable AI compliance gain competitive advantage in regulatory and investor relations.

11. Risk & Mitigation

Risk	Severity	Mitigation	Residual Risk
LLM hallucination in narrative	**HIGH**	Evidence-bounded input; temp 0.3; 8-check post-generation validation; explainability scoring; mandatory analyst review	LOW — multi-layer defence
PII leakage to LLM	**HIGH**	Architectural isolation in Evidence Pack Builder — not a config setting; lossy one-way transformation enforced by design	VERY LOW — by architecture
AML rule gaps (novel typologies)	**MEDIUM**	Git-versioned rule library with governed update process; analyst rejection feedback loop; typology drift monitoring	MEDIUM — process dependent
Model drift over time	**MEDIUM**	5 continuous KPI monitors; validation failure rate threshold alerts; monthly corpus and prompt review cadence	LOW — monitored
Corpus quality degradation	**MEDIUM**	Version-controlled corpus; snapshot IDs logged per retrieval; retrieval relevance score tracked over time	LOW — monitored
LLM inference unavailability	**LOW**	Max 2 regeneration retries with timeout; circuit breaker; cases	VERY LOW

		escalated to manual drafting with evidence pack pre-populated	
Audit log integrity breach	**HIGH**	Append-only PostgreSQL design; AES-256 encryption at rest; no delete permissions at application layer; SIEM integration	VERY LOW — by design
Prompt injection attack	**MEDIUM**	Jinja2 strict input sanitisation; system instructions immutable at runtime; all injection attempts logged as security events	LOW — sanitised

12. Deployment Models

Model	Architecture	Best For	Key Trade-Off
On-Premises (Air-Gapped)	Full local stack: Ollama for LLM inference, self-hosted PostgreSQL and ChromaDB, Docker Compose	Strict data residency requirements; regulatory mandates prohibiting cloud processing	Higher infra cost; manual scaling; no managed services
Private Cloud (AWS / Azure / GCP)	Kubernetes HPA; dedicated GPU instances for vLLM; managed PostgreSQL, Redis, and object storage	Institutions requiring elasticity and managed operational overhead	Data sovereignty controls must be configured; higher ongoing spend
Hybrid	Ingestion, enrichment, and rule engine on-premises; LLM inference in private cloud tenancy with network isolation	Institutions balancing data control with GPU elasticity for high-volume periods	Network latency between layers; dual-environment operational complexity

13. Live Demo Flow

The demonstration follows a structuring-and-layering typology case. A judge can follow the complete journey from raw alert to filed SAR with full audit trail in under ten minutes. All services run locally via Docker Compose — no internet connection required.

#	Story Beat	What the Judge Sees
1	**The Alert Arrives**	JSON alert submitted via Postman/cURL. Case ID generated instantly. Dashboard shows INGESTED state.
2	**Intelligence Activates**	Real-time state transitions: ENRICHED → RULE_TRIGGERED within ~10 seconds. Risk score appears on dashboard.
3	**Rules Fire**	Evidence panel populates: AML-001 [Structuring, 0.82] and AML-017 [Rapid Movement, 0.79] with specific observations.

4	**AI Retrieves Precedents**	ChromaDB returns top-5 matched SARs with similarity scores. Corpus snapshot ID logged.
5	**Narrative Generated**	LLM produces structured JSON in ~90 seconds. Draft populates into schema sections in analyst UI.
6	**Validation Passes**	All 8 validation checks pass. Explainability scores shown per sentence. No sentences flagged.
7	**Analyst Reviews**	Side-by-side evidence panel and draft editor. Hover over sentence → evidence links shown. Clean, professional UI.
8	**Edit → Approve**	Analyst modifies one sentence. Edit captured as section diff. Approval submitted with mandatory comment.
9	**Audit Trail Revealed**	Full audit record displayed: rules → evidence → corpus snapshot → prompt SHA → generation → validation → edit diff → approval.
10	**Replay Demonstrated**	Case ID entered into replay tool. System regenerates identical draft from stored state. Judge sees exact reproducibility.

14. Future Roadmap

Horizon	Timeline	Enhancement
SHORT TERM	**Q2 2026**	Full Kafka streaming integration — sub-minute latency from alert to draft availability at enterprise volume
	Q3 2026	Statistical explainability scoring — per-sentence confidence quantification with configurable acceptance thresholds
	Q3 2026	Cross-border compliance engine — jurisdiction-aware rule sets for FinCEN, FCA, RBI, MAS simultaneous compliance
MID TERM	**Q4 2026**	Graph intelligence layer (Neo4j) — multi-hop entity relationship analysis detecting circular flows and beneficial ownership structures
	Q1 2027	Adaptive rule learning — analyst feedback loop proposing YAML rule candidates from approval/rejection patterns for compliance team review
	Q1 2027	Domain-fine-tuned LLM — compliance-specific model training on approved SAR corpus for superior typology accuracy beyond RAG alone
ENTERPRISE	**2027+**	Typology drift detection — statistical monitoring of rule trigger frequency to surface emerging AML patterns before enforcement exposure
	2027+	Multilingual narrative generation — jurisdiction-appropriate SAR formatting in multiple languages for global compliance operations

15. Why Our Solution Wins

DIRECT COMPETITIVE COMPARISON — 8 CRITICAL COMPLIANCE CAPABILITIES

Capability	Manual Process	Generic LLM	Rule-Only AML	This System
Evidence-grounded SAR narrative	◐	✗	✗	✓
Deterministic AML rule enforcement	◐	✗	✓	✓ versioned
Sentence-level audit traceability	✗	✗	✗	✓
PII isolated from AI model	N/A	✗	N/A	✓ by design
Regulatory schema enforcement	◐	✗	✗	✓
Mandatory human approval governance	✓	✗	✗	✓
Regulator-replayable complete audit	✗	✗	◐	✓
Multi-layer hallucination controls	N/A	✗	N/A	✓ 3 layers

What Competitors Cannot Match	Why It Matters to Regulators and Banks
Architectural PII isolation	Not a filter or policy — it is a physical data boundary. The LLM literally cannot access PII. This is not achievable by fine-tuning or prompting alone.
Deterministic decision layer	Compliance determinations are reproducible, versionable, and jurisdiction-configurable. They can be reviewed by regulators and updated through a governed process.
Regulator-replayable audit	No other approach can regenerate the exact same output from stored state. This capability is the difference between 'we used AI' and 'we can prove what our AI did'.
Evidence-bounded generation	The LLM operates in a closed information space. It cannot speculate. Every claim is traceable. This converts AI assistance from a liability into a regulatory asset.

16. Prototype Demonstration

A functional prototype of the SAR Narrative Generator has been successfully developed and tested locally.
The system performs end-to-end processing including:

- Risk scoring
- Evidence analysis
- AI-generated SAR narrative
- Structured regulatory report formatting
- PDF report generation

Due to round-one submission constraints, the prototype is demonstrated through a recorded walkthrough instead of a live deployment.

Demo Video:

https://drive.google.com/file/d/1as51ormW3jyvQZpLJVuaG104Z0Tlp9_S/view?usp=sharing

Git Link:

https://github.com/rohit90966/Hacakthon-B.git

This video showcases the full workflow from input data ingestion to final Suspicious Activity Report generation.

17. Conclusion

The SAR Narrative Generator with Intelligent Audit Trail is not an AI writing tool applied to compliance. It is a compliance system that uses AI precisely where AI adds value — and excludes it precisely where it introduces risk.

The system's four defining innovations — architectural PII isolation, deterministic rule-layer separation, sentence-level evidence traceability, and regulator-replayable audit replay — are not features layered on top of an LLM. They are the structural foundation upon which the LLM operates. Remove any one of them and the system loses its regulatory credibility. Together, they transform AI assistance from a compliance liability into a demonstrable compliance asset.

For a bank processing 10,000 SARs annually, the system delivers an estimated £1.8 million net annual saving while improving narrative quality, reducing regulatory exposure, and generating the complete audit evidence that modern supervisors demand. At 50,000 SARs, the saving scales toward

£9 million. The architecture scales with it — every component is stateless, independently scalable, and production-deployable from the prototype foundation built here.

In a regulatory environment that is rapidly evolving toward explicit AI accountability requirements — FCA, EBA, and the EU AI Act all moving toward mandatory explainability for high-risk financial AI — this system is not just compliant with those requirements. It is designed to exemplify them.

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